Rotaract Club of Xiamen University Malaysia | Rotary Club of Kuala Lumpur Dynamics | Rotary Club of Tun Razak Exchange

SDC Hackathon

Technical Tools Declaration

Hosted by Xiamen University Malaysia (XMUM) AI Club
Submission Deadline: 25 October 2026

Instructions: Declare every API, cloud service, AI tool, dataset, model, and sponsor resource your team used to build the project. Complete every section; where a category does not apply, tick "No" or write "None" – do not leave it blank. Submit as a single PDF document named "SDC2026_TeamName_TechDeclaration.pdf" together with your project submission. Undeclared third-party resources discovered at judging may affect your score or eligibility.

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SDC HACKATHON 2026

Technical & Tools Declaration

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1. Team & Project Overview

Team Name	AlgorithmX
Project / Solution Name	AI Skill Navigator
Name (Leader)	CHEN ZANXU
Email & Phone (Leader)	chenzanxu430@gmail.com 601135189975
University Affiliation(s)	Xiamen University Malaysia
Live Prototype URL	https://algorithm-xsdc-1.replit.app
Code Repository URL (public / access granted to judges)	https://github.com/chenzanxu430-code/AlgorithmX_SDC

2. Architecture & Technology Stack

2.1 System Architecture Overview

Describe how the components declared in this document connect end-to-end (e.g., user interface → backend → AI model/API → database → cloud hosting). Attach a system block diagram as an appendix if available.

[Maximum 200 words]

The prototype is a web application hosted on Replit. Its React frontend collects users' role, tasks, experience, interests, and skills, then uses Axios to communicate with backend endpoints. The backend runs on Node.js and uses Express to provide application APIs and process submitted information through rule-based analysis. Mongoose provides the data-access layer for MongoDB, and dotenv loads environment configuration. The application combines curated role and skill profiles with labour-market trend data to generate skill-gap insights, adjacent-role recommendations, and a learning plan. Results are returned to the frontend for display. The prototype does not call an external GenAI model or third-party runtime API.
System flow: User → React frontend → Axios → Express/Node.js backend → rule-based analysis and MongoDB via Mongoose → results displayed in the React frontend; hosted on Replit.

2.2 Languages, Frameworks & Libraries

List the main programming languages, frameworks, and third-party libraries that your team did not create. Open-source use is permitted provided it is declared and properly licensed/attributed.

Name & Version	Purpose in Project	License
Node.js 18.17.0.	JavaScript runtime for running the server-side application	MIT
Express 4.18.2	Backend web framework for implementing application endpoints.	MIT
React 18.2.0	Frontend library for building the user interface.	MIT
Axios 1.6.0	HTTP client for sending and receiving application data.	MIT
Mongoose 7.6.3	Object data modeling library for connecting the Node.js application to MongoDB.	MIT

Name & Version	Purpose in Project	License
dotenv 16.3.1	Loads configuration values from environment files into the application environment.	BSD-2-Clause

3. APIs & Cloud Services

3.1 External APIs

List every external API your solution calls (GenAI, maps, payments, messaging, data feeds, etc.). Do not include API keys or secrets in this document.

API & Provider	Purpose in Project	Access Tier (free / paid / sponsor-provided)
None	No external runtime APIs are called.	N/A
None	No external runtime APIs are called.	N/A
None	No external runtime APIs are called.	N/A

3.2 Cloud Platforms & Deployment

List every cloud service used for hosting, storage, databases, authentication, or compute, and indicate how the prototype is deployed.

Service & Provider	Purpose in Project	Plan / Region
Replit	Development environment and hosting/deployment of the live prototype.	Replit Core$20.00 / month/Malaysia
MongoDB	Database accessed by the Node.js backend through Mongoose.	Not specified

4. AI Tools, Models & Datasets

4.1 AI Models Used in the Solution

List every AI model embedded in or called by your application, including pre-trained, fine-tuned, and self-trained models.

Model & Version	Source / Provider	Usage (pre-trained / fine-tuned / self-trained)	License / Terms
None	N/A	N/A	N/A

4.2 Datasets

List all datasets used for training, fine-tuning, evaluation, or in the running application. State clearly if a dataset contains personal or sensitive data.

Dataset Name	Source / URL	Type (public / private / self-collected / synthetic)	License	Personal Data? (Y/N)
Malaysia JobStreet job-posting collection: e-commerce operations, data analysis, AI/content, and related roles	Multiple public JobStreet listing URLs; see the complete URL inventory in the attached dataset list or project docs/ folder.	Self-collected from public listings	No separate dataset license identified; source terms apply.	N
The Future of Jobs Report 2025	https://www.weforum.org/publications/the-future-of-jobs-report-2025/	Public report	No separate dataset license identified; cite the source and follow its terms.	N
Labour Force Statistics, August 2026	https://www.dosm.gov.my/portal-main/release-content/labour-force-statistics-aug2026	Public official statistics	No separate dataset license identified in the inventory; follow DOSM's terms.	N

Dataset Name	Source / URL	Type (public / private / self-collected / synthetic)	License	Personal Data? (Y/N)
Malaysia Digital Economy Blueprint	https://www.mydigital.gov.my/wp-content/uploads/2023/08/Malaysia-Digital-Economy-Blueprint_ENG.pdf	Public policy document	No separate dataset license identified; cite the source and follow its terms.	N

4.3 AI-Assisted Development Tools

Tools used to help write code, design, content, or documentation (e.g., chat assistants, code completion tools, image generators). Declaring these tools does not reduce your score; undeclared use may.

Did your team use any generative-AI tool to help create code, design, text, or other project material?

☑ Yes ☐ No

If yes, list them below:

Tool & Version	**Used For**	**Extent of Use (minor / moderate / major)**
Replit Agent	Assisted in adapting the existing Next.js application for Replit's pnpm workspace and resolving setup and deployment issues. It was used during development and is not called by the running application.	Moderate
ChatGPT	Used as an AI-assisted research, synthesis, and editorial tool to structure and refine the project's problem framing, solution narrative, technical explanations, and competition submission materials. All AI-assisted content was reviewed, fact-checked, and adapted by the team to reflect the actual implementation.	Moderate

Briefly describe how your team reviewed, tested, and took responsibility for AI-generated output:

[Maximum 100 words]

Team members reviewed AI-assisted code changes and written content against the existing application, project information, and submission requirements. The application was checked in Replit's preview and published environment. Suggestions that did not match the project or its actual implementation were corrected or omitted. No generative-AI model is called by the running application.

5. Sponsor-Provided Resources

Declare any official SDC sponsor-provided GenAI API, cloud credits, tools, mentoring, or data that your team used.

Did your project use any official SDC sponsor-provided resource?

☑Yes ☐No

If yes, list them below:

Sponsor	**Resource (API / credits / platform / data)**	**How It Was Used in the Project**
Replit	Development platform and cloud deployment	The prototype was developed and deployed on Replit.

Was the sponsor resource essential to the core functionality of the prototype, or could it be replaced? Explain briefly:

[Maximum 100 words]

Replit was used for development and hosting. The application's rule-based analysis does not depend on Replit-specific AI services and could be deployed on another compatible platform.

6. Security, Privacy & Responsible Use

6.1 Data Handling & Privacy

What user or personal data does your application collect, store, or send to third-party services (including AI APIs)? How is it protected, and how long is it kept? State "No user data is collected" if applicable.

[Maximum 150 words]

The prototype does not require users to log in. Users provide their current role, daily tasks, work experience, interests, and skills. The application processes this information to generate a personalized skills analysis and career recommendations. A saved analysis session is stored in the user's browser so the result can be reopened; it remains there until the user clears the browser's site data or replaces the saved session. The application does not request names, email addresses, or other direct identifiers. User profile information is not sent to an external GenAI model or third-party API. The application's own trend-data service provides workforce trend information and does not receive the user's profile.

6.2 Known Limitations & Risks

Identify the main technical limitations of your AI components (e.g., hallucination, bias, rate limits, dependency on a single provider) and any mitigation you have implemented.

[Maximum 150 words]

The prototype provides directional, rule-based estimates based on curated role and skill profiles and a limited set of manually reviewed job-posting snapshots. The data may not represent the whole Malaysian labour market and may reflect source or sampling bias. Job requirements can also change after the snapshots are collected. Role-match scores are not hiring probabilities, and recommendations do not guarantee employment. The prototype displays source dates, sample sizes and explanatory factors to help users interpret results. Broader role coverage and user testing are planned for Phase 2.

7. Compliance & Readiness Checklist

Tick all boxes that apply before submitting.

- ☑ All APIs, cloud services, AI tools, models, and datasets used in the project are declared in this document.
- ☑ All third-party code, datasets, and models are used in accordance with their licenses, and attribution is provided where required.
- ☑ No API keys, passwords, or other credentials are exposed in the code repository, prototype, or this document.
- ☑ Any use of generative-AI tools for development has been disclosed and reviewed by the team.
- ☑ Any use of official SDC sponsor resources has been declared in Section 5.
- ☑ The prototype described here is accessible to judges through the links provided in Section 1.
- ☑ All fields in this template have been completed within the word limits stated.

8. Declaration

By submitting this document, the team leader confirms that the information provided is complete and accurate to the best of the team's knowledge, and that every API, cloud service, AI tool, dataset, model, and sponsor resource used in the project has been declared.



[Signature]

[Date]10/24/2026